Lab 6: Nexus Repository and User Access

Course	CS-649: Cloud DevOps
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Repository / Directory	DevOps / Labs/Lab06

1. Task 1: Nexus container running with ports 8006 and 8007

docker ps | grep nexus shows the running nexus container with published ports 8006 and 8007.

nexus_container_running.png

```
amnapervez8910@ubuntu:~$ docker ps | grep nexus
3d00c7bfda4e   sonatype/nexus3:latest          "/opt/sonatype/nexus…"   8 hours ago   Up 55 minutes   0.0.0.0:8007->8007/tcp, [::]:8007
7/tcp, 0.0.0.0:8006->8081/tcp, [::]:8006->8081/tcp   nexus
amnapervez8910@ubuntu:~$
```

2. Task 2: nexus user verified through Status API (HTTP 200)

curl -i -u nexus against /service/rest/v1/status returns HTTP/1.1 200 OK; the password is not visible.

nexus_user_access.png

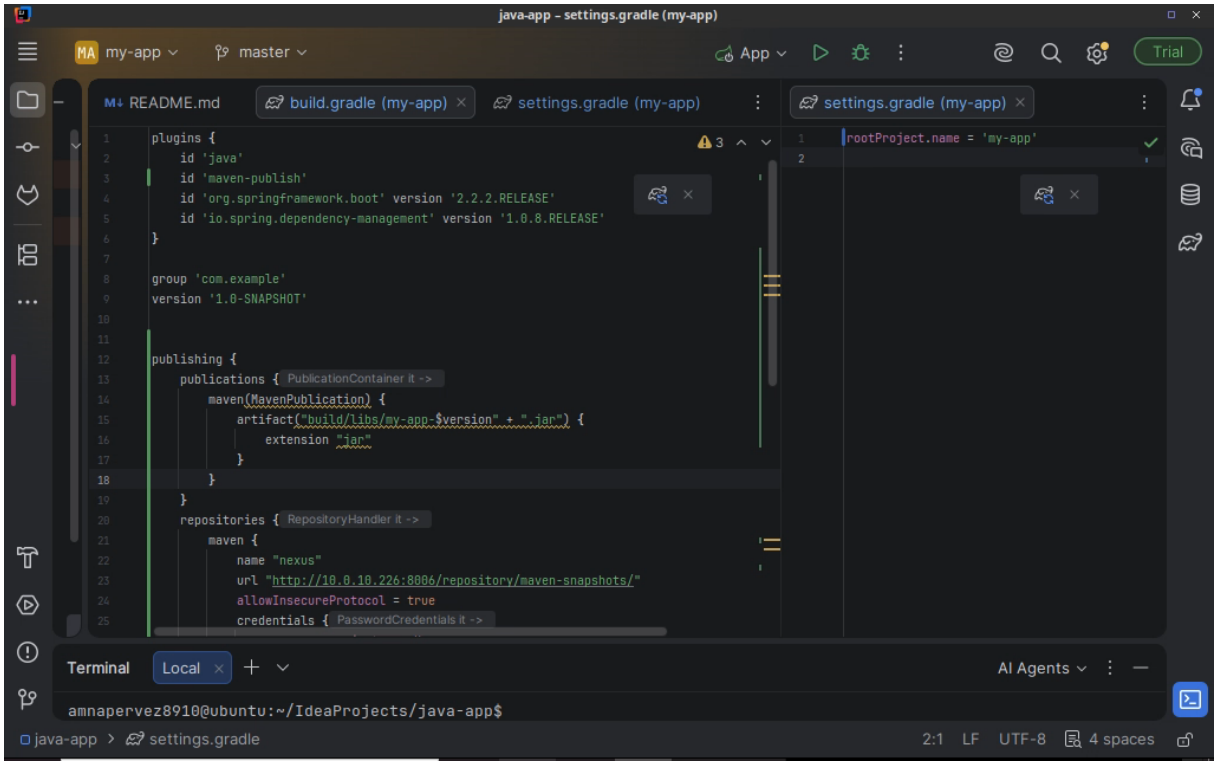
```
amnapervez8910@ubuntu:~$ curl -i -u nexus http://localhost:8006/service/rest/v1/status
Enter host password for user 'nexus':
HTTP/1.1 200 OK
Server: Nexus/3.98.1-01 (COMMUNITY)
Vary: Accept-Encoding
X-Content-Type-Options: nosniff
Content-Security-Policy: default-src http: data: blob: 'unsafe-inline'; script-src http: 'unsafe-inline' 'unsafe-eval'
Content-Length: 0

amnapervez8910@ubuntu:~$
```

3. Task 3: Gradle build.gradle and settings.gradle configuration

build.gradle (java, maven-publish, publishing block, Nexus URL) and settings.gradle with rootProject.name = 'my-app'.

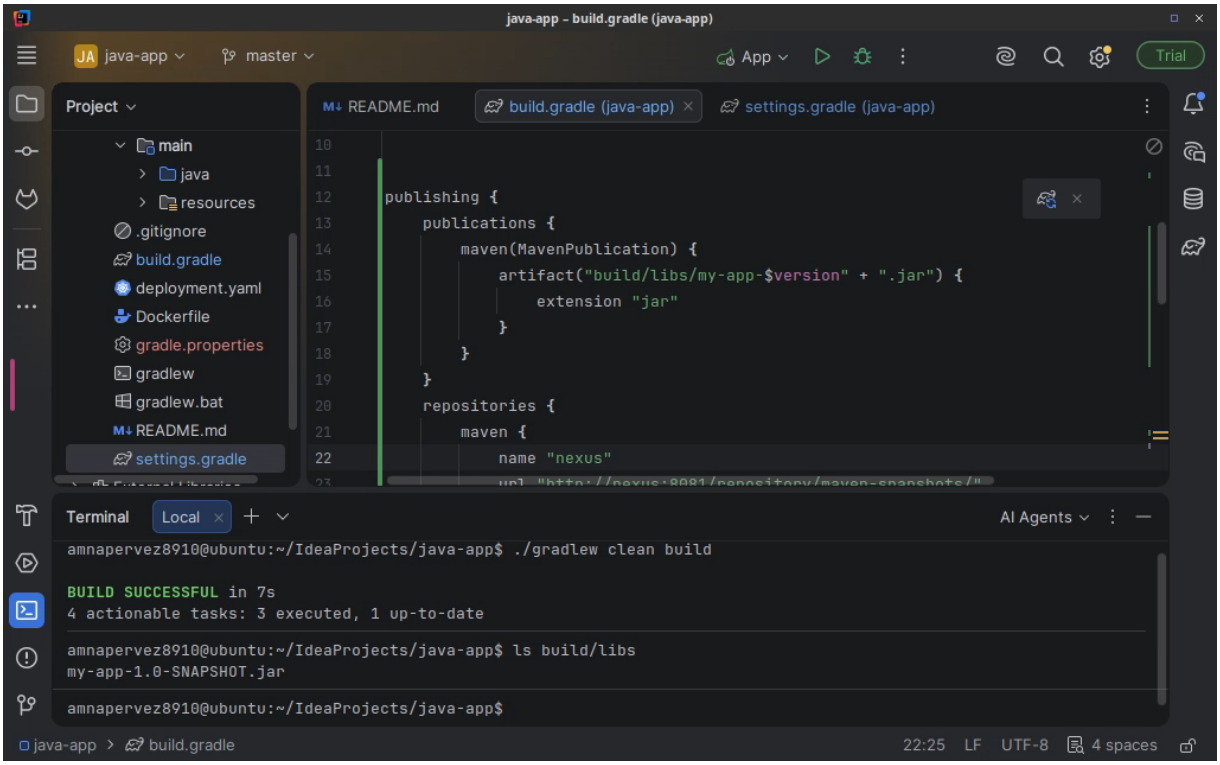
gradle_nexus_configuration.png



4. Task 3: ./gradlew clean build, BUILD SUCCESSFUL and my-app-1.0-SNAPSHOT.jar

The Gradle build finishes with BUILD SUCCESSFUL and creates my-app-1.0-SNAPSHOT.jar in build/libs.

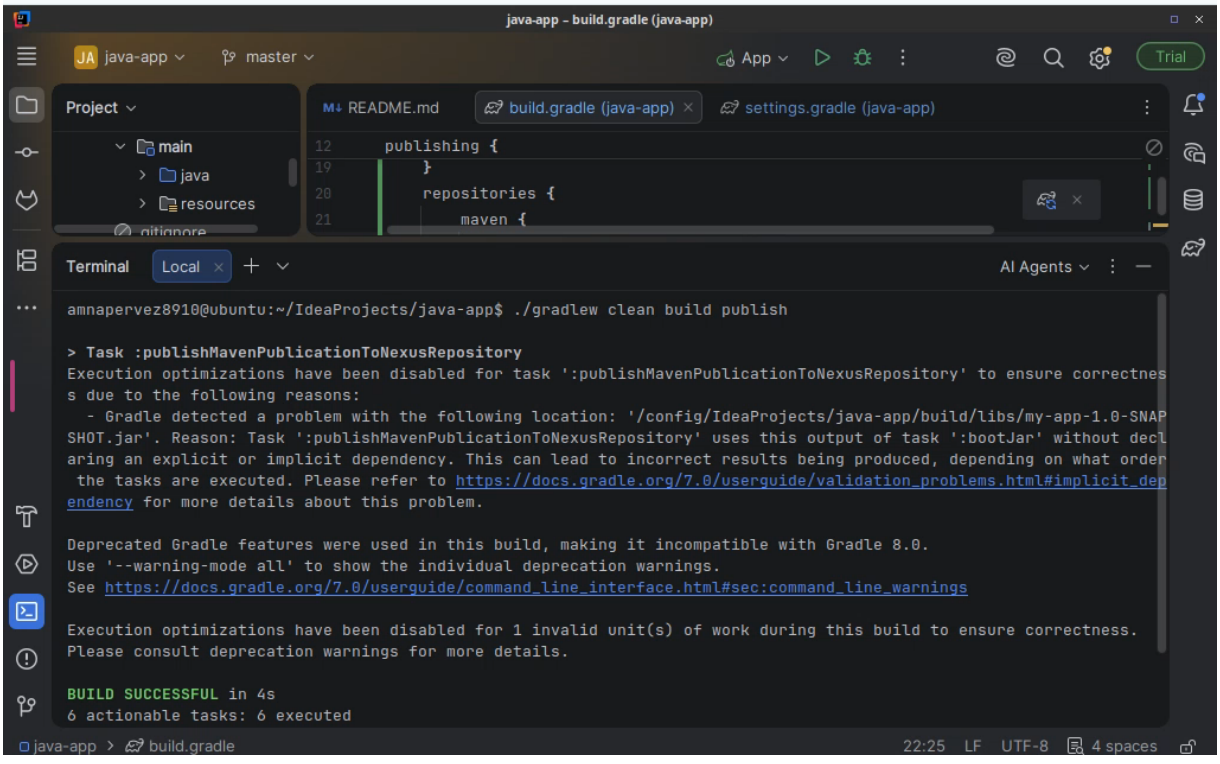
gradle_nexus_build.png



5. Task 3: ./gradlew clean build publish, BUILD SUCCESSFUL

publishMavenPublicationToNexusRepository runs and the build finishes with BUILD SUCCESSFUL.

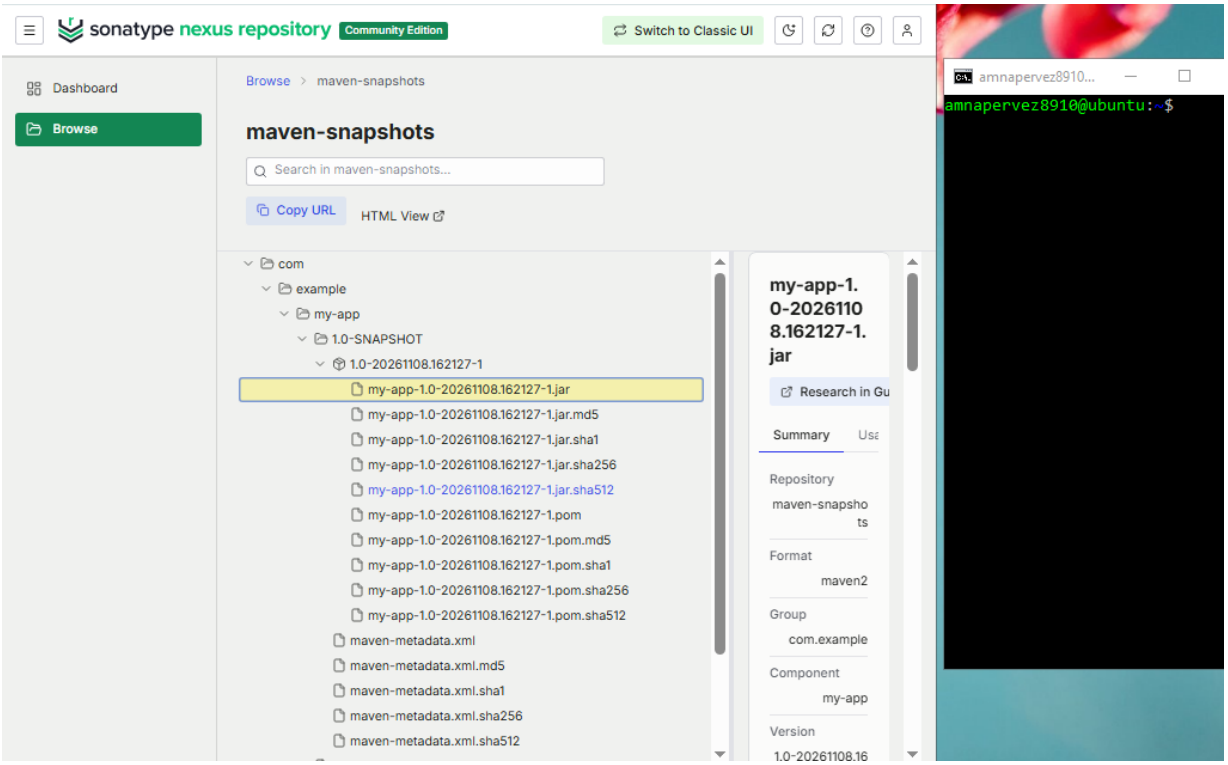
gradle_nexus_publish.png



6. Task 3: Gradle JAR and metadata in Nexus maven-snapshots

maven-snapshots contains com/example/my-app/1.0-SNAPSHOT with the JAR, POM, checksums and maven-metadata.xml.

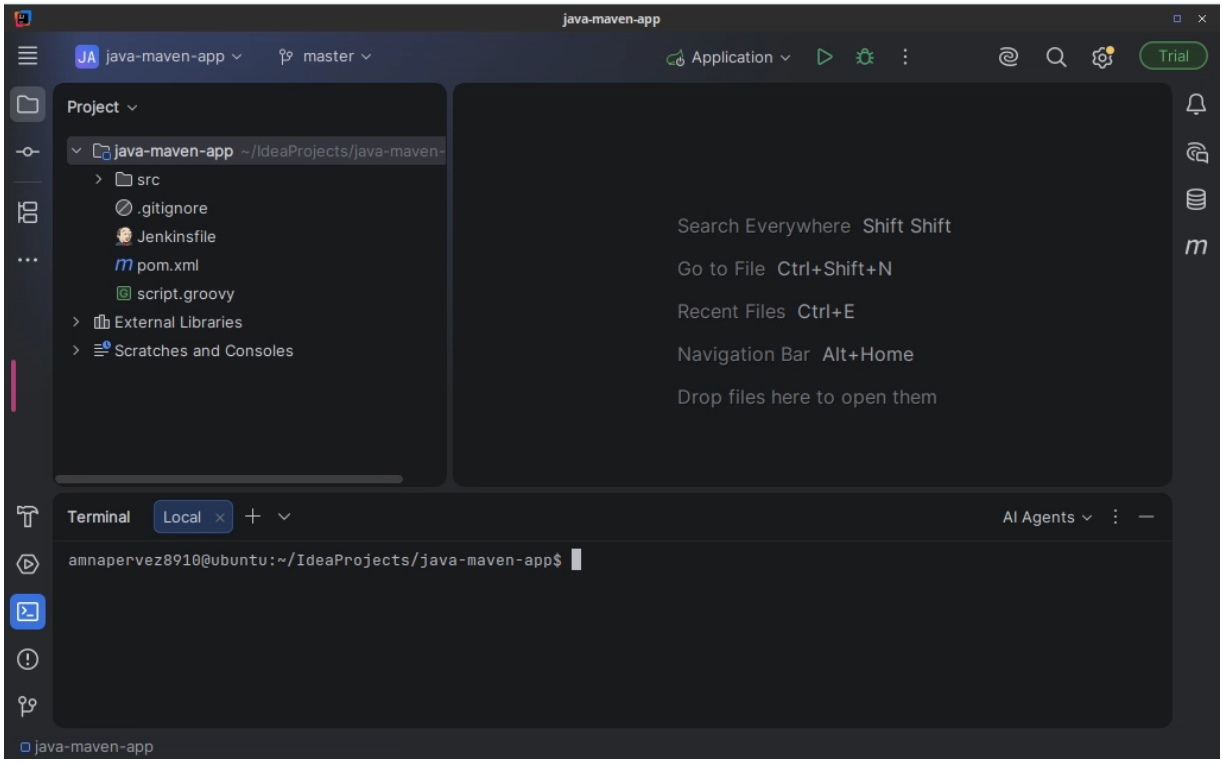
nexus_gradle_artifact.png



7. Task 4: java-maven-app opened in IntelliJ

The java-maven-app project is open in IntelliJ IDEA.

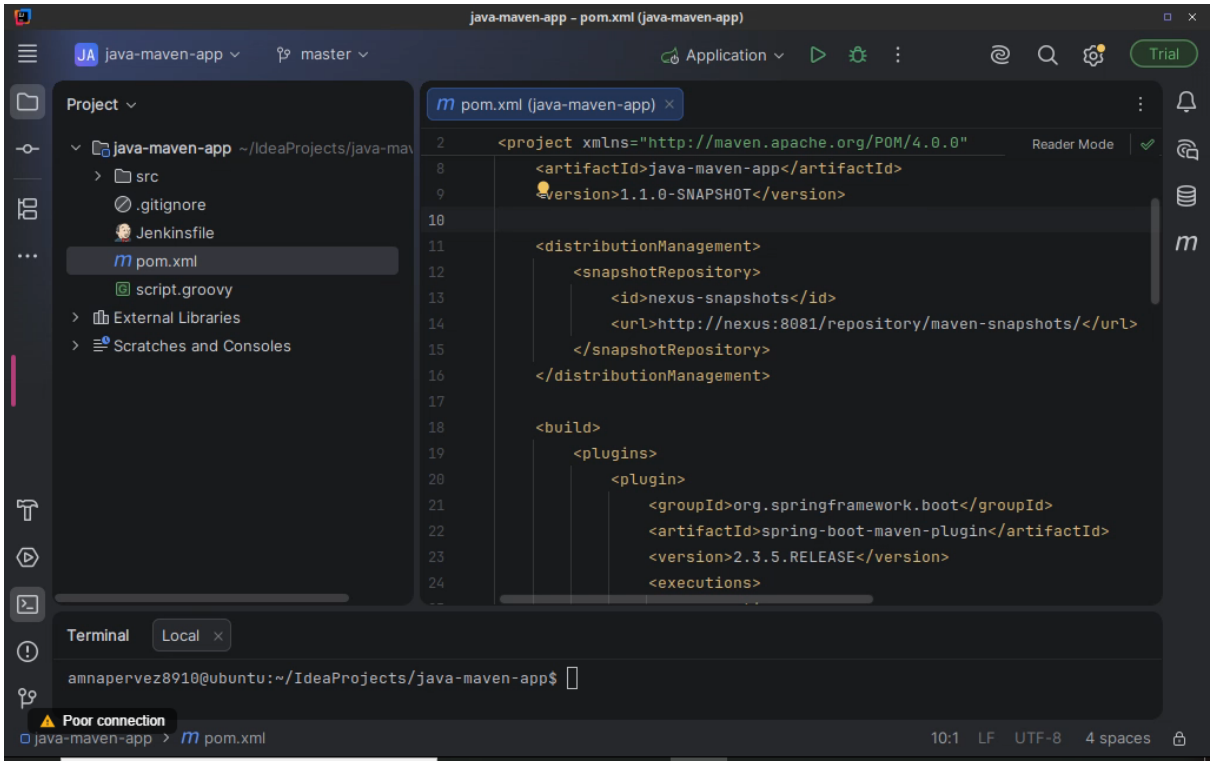
maven_nexus_project.png



8. Task 4: distributionManagement block in pom.xml

The saved distributionManagement block with snapshotRepository id nexus-snapshots in pom.xml.

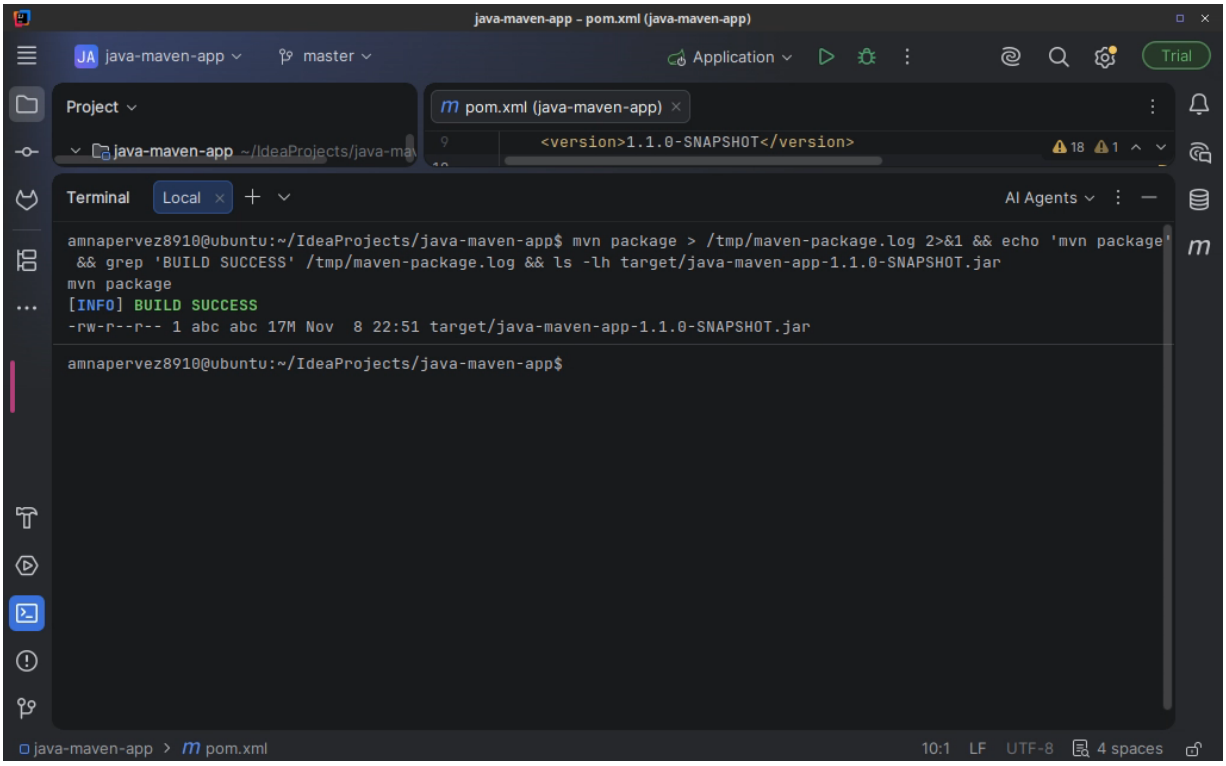
maven_distribution_management.png



9. Task 4: mvn package, BUILD SUCCESS and generated JAR

mvn package finishes with BUILD SUCCESS and creates target/java-maven-app-1.1.0-SNAPSHOT.jar.

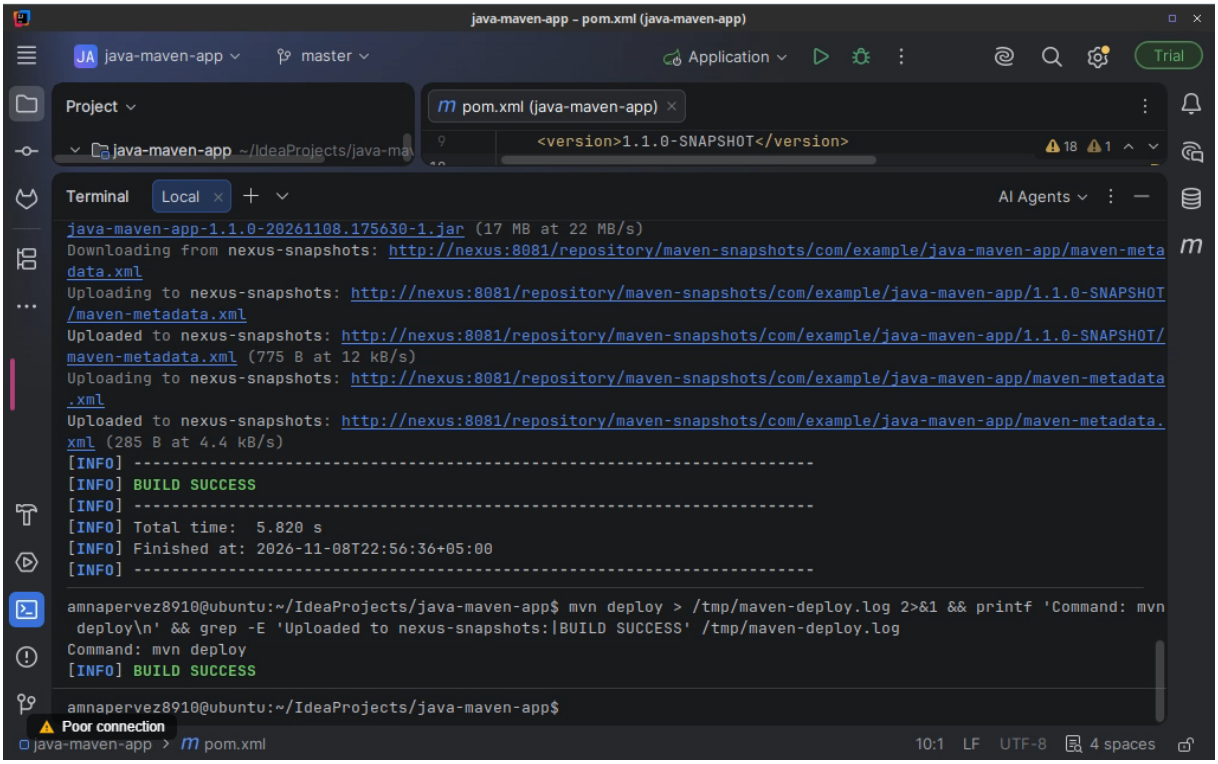
maven_package_nexus.png



10. Task 4: mvn deploy, upload lines and BUILD SUCCESS

mvn deploy uploads the JAR, POM and metadata to nexus-snapshots and ends with BUILD SUCCESS.

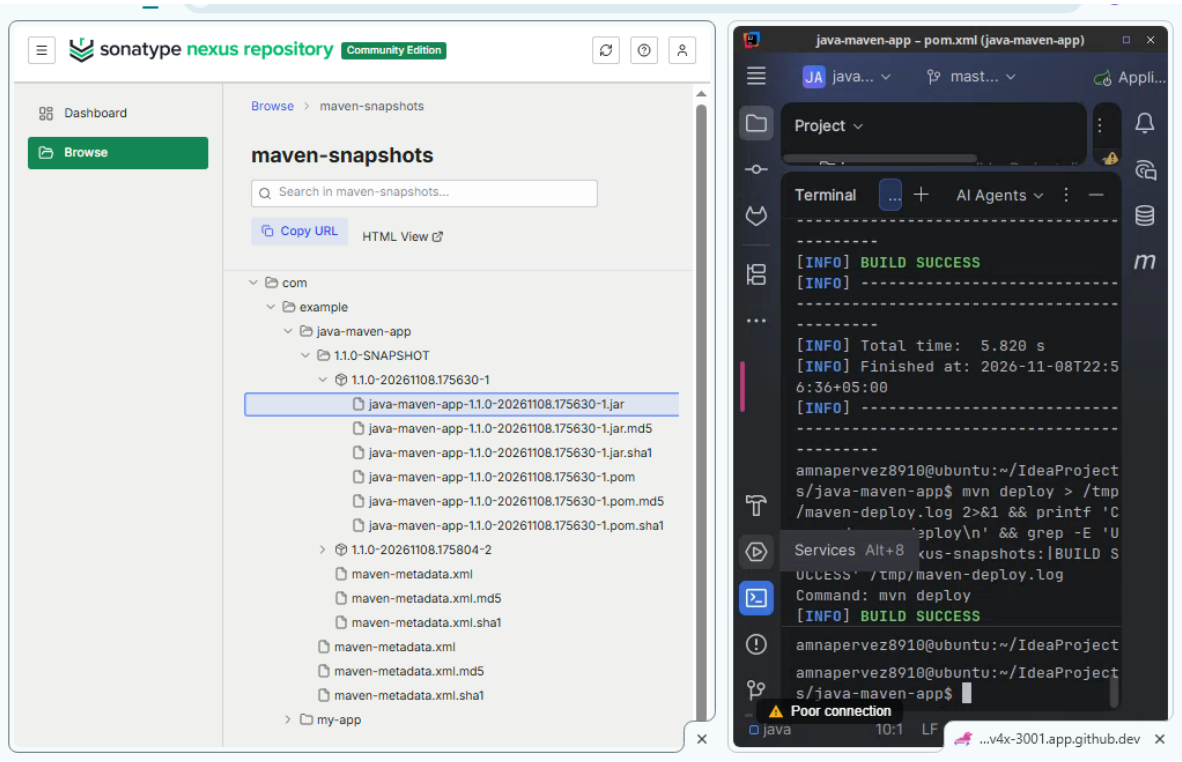
maven_deploy_nexus.png



11. Task 4: Maven JAR, POM and metadata in Nexus

maven-snapshots contains com/example/java-maven-app/1.1.0-SNAPSHOT with the JAR, POM and metadata.

nexus_maven_artifact.png



12. Task 5: repository list as nexus user

REST API repository list returned for the restricted nexus user.

nexus_repositories_user.png

```
amnapervez8910@ubuntu:~$ curl -u nexus -X GET "http://localhost:8006/service/rest/v1/repositories"
Enter host password for user 'nexus':
[ {
  "name" : "maven-snapshots",
  "format" : "maven2",
  "type" : "hosted",
  "url" : "http://localhost:8006/repository/maven-snapshots",
  "online" : true,
  "size" : 0,
  "attributes" : { }
} ]amnapervez8910@ubuntu:~$
```

13. Task 5: repository list as admin user

REST API repository list returned for the admin user (more repositories visible).

nexus_repositories_admin.png

```
amnapervez8910@ubuntu:~$ curl -u admin -X GET "http://localhost:8006/service/rest/v1/repositories"
}, {
  "name" : "nuget-hosted",
  "format" : "nuget",
  "type" : "hosted",
  "url" : "http://localhost:8006/repository/nuget-hosted",
  "online" : true,
  "size" : 0,
  "attributes" : { }
}, {
  "name" : "nuget.org-proxy",
  "format" : "nuget",
  "type" : "proxy",
  "url" : "http://localhost:8006/repository/nuget.org-proxy",
  "online" : true,
  "size" : 0,
  "attributes" : {
    "proxy" : {
      "remoteUrl" : "https://api.nuget.org/v3/index.json"
    }
  }
}, {
  "name" : "nuget-group",
  "format" : "nuget",
  "type" : "group",
  "url" : "http://localhost:8006/repository/nuget-group",
  "online" : true,
  "size" : 0,
  "attributes" : { }
} ]amnapervez8910@ubuntu:~$
```

14. Task 5: components in maven-snapshots

Components in maven-snapshots with their id, repository, name and version.

nexus_components_list.png

```
amnapervez8910@ubuntu:~$ curl -sS -u nexus -X GET "http://localhost:8006/service/rest/v1/components?repository=maven-snapshots" | jq
'.items[] | {id, repository, name, version}'
Enter host password for user 'nexus':
{
  "id": "bWF2ZW4tc25hcHNob3RzOjRmMWJiY2Rk",
  "repository": "maven-snapshots",
  "name": "my-app",
  "version": "1.0-20261108.162127-1"
}
{
  "id": "bWF2ZW4tc25hcHNob3RzOjllMzc3OWJh",
  "repository": "maven-snapshots",
  "name": "java-maven-app",
  "version": "1.1.0-20261108.175630-1"
}
{
  "id": "bWF2ZW4tc25hcHNob3RzOmVkNTMzNjk3",
  "repository": "maven-snapshots",
  "name": "java-maven-app",
  "version": "1.1.0-20261108.175804-2"
}
amnapervez8910@ubuntu:~$
```

15. Task 5: component details by ID

One component retrieved by its ID, showing repository, format, group, name, version and assets.

nexus_component_details.png

```
amnapervez8910@ubuntu:~$ curl -sS -u nexus "http://localhost:8006/service/rest/v1/components/$ID" | jq '{id, repository, format, grou
p, name, version, assets: [.assets[].path]}'
Enter host password for user 'nexus':
{
  "id": "bWF2ZW4tc25hcHNob3RzOmVkNTMzNjk3",
  "repository": "maven-snapshots",
  "format": "maven2",
  "group": "com.example",
  "name": "java-maven-app",
  "version": "1.1.0-20261108.175804-2",
  "assets": [
    "/com/example/java-maven-app/1.1.0-SNAPSHOT/java-maven-app-1.1.0-20261108.175804-2.pom",
    "/com/example/java-maven-app/1.1.0-SNAPSHOT/java-maven-app-1.1.0-20261108.175804-2.pom.sha1",
    "/com/example/java-maven-app/1.1.0-SNAPSHOT/java-maven-app-1.1.0-20261108.175804-2.pom.md5",
    "/com/example/java-maven-app/1.1.0-SNAPSHOT/java-maven-app-1.1.0-20261108.175804-2.jar",
    "/com/example/java-maven-app/1.1.0-SNAPSHOT/java-maven-app-1.1.0-20261108.175804-2.jar.sha1",
    "/com/example/java-maven-app/1.1.0-SNAPSHOT/java-maven-app-1.1.0-20261108.175804-2.jar.md5"
  ]
}
amnapervez8910@ubuntu:~$
```

16. Task 6: default blob store content inside the container

docker exec into the nexus container; blobs directory lists default and its content.

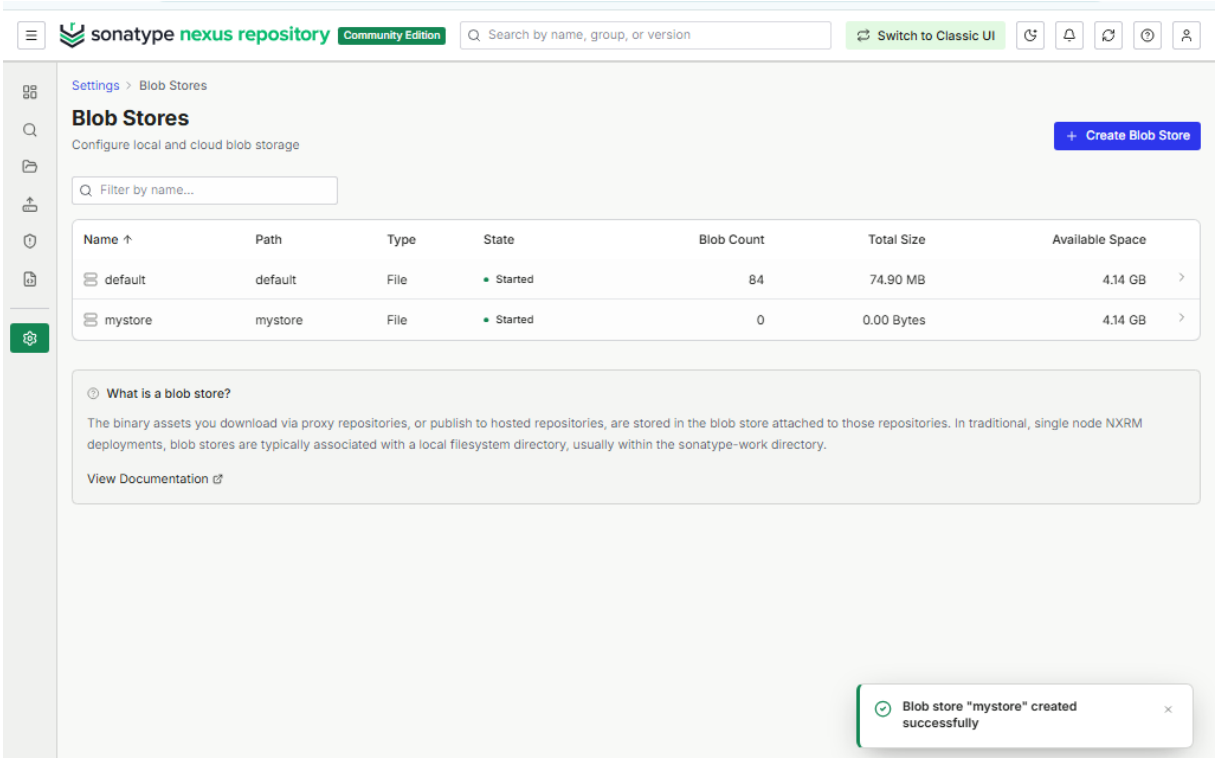
nexus_default_blob_content.png

```
amnapervez8910@ubuntu:~$ docker exec -it nexus /bin/sh
/opt/sonatype $ cd sonatype-work/nexus3
/opt/sonatype/sonatype-work/nexus3 $ cd blobs && ls && ls default/content
default
2026        directpath  tmp
/opt/sonatype/sonatype-work/nexus3/blobs $
```

17. Task 6: mystore file blob store created

Blob Stores page shows default and mystore, type File, state Started.

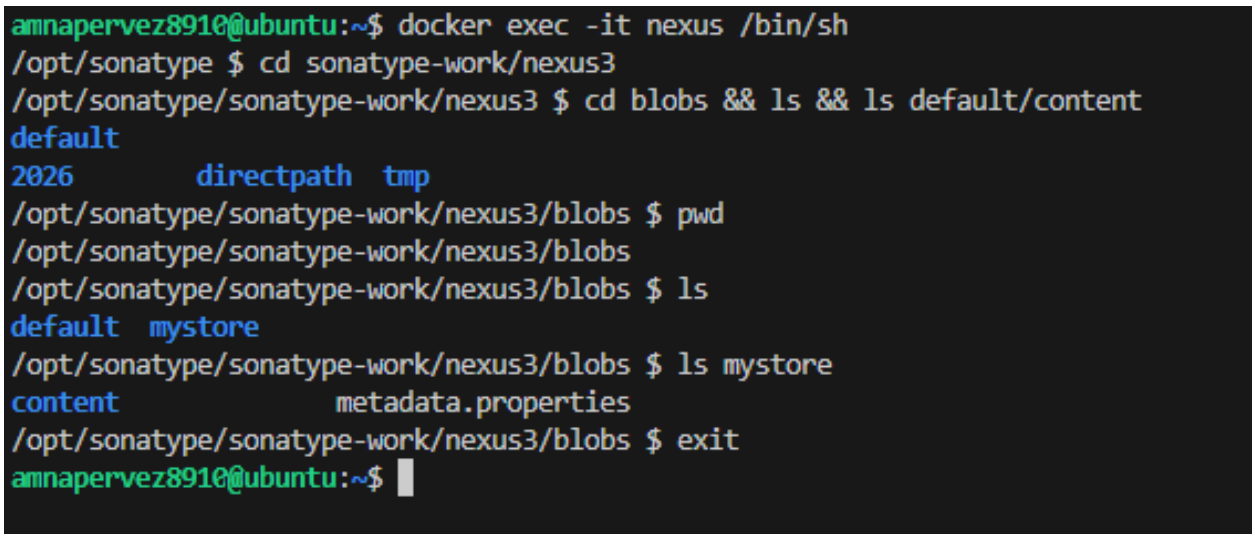
nexus_mystore_created.png



18. Task 6: blob directories (default and mystore)

Inside the container: pwd shows blobs, ls shows default and mystore, then exit returns to the Ubuntu prompt.

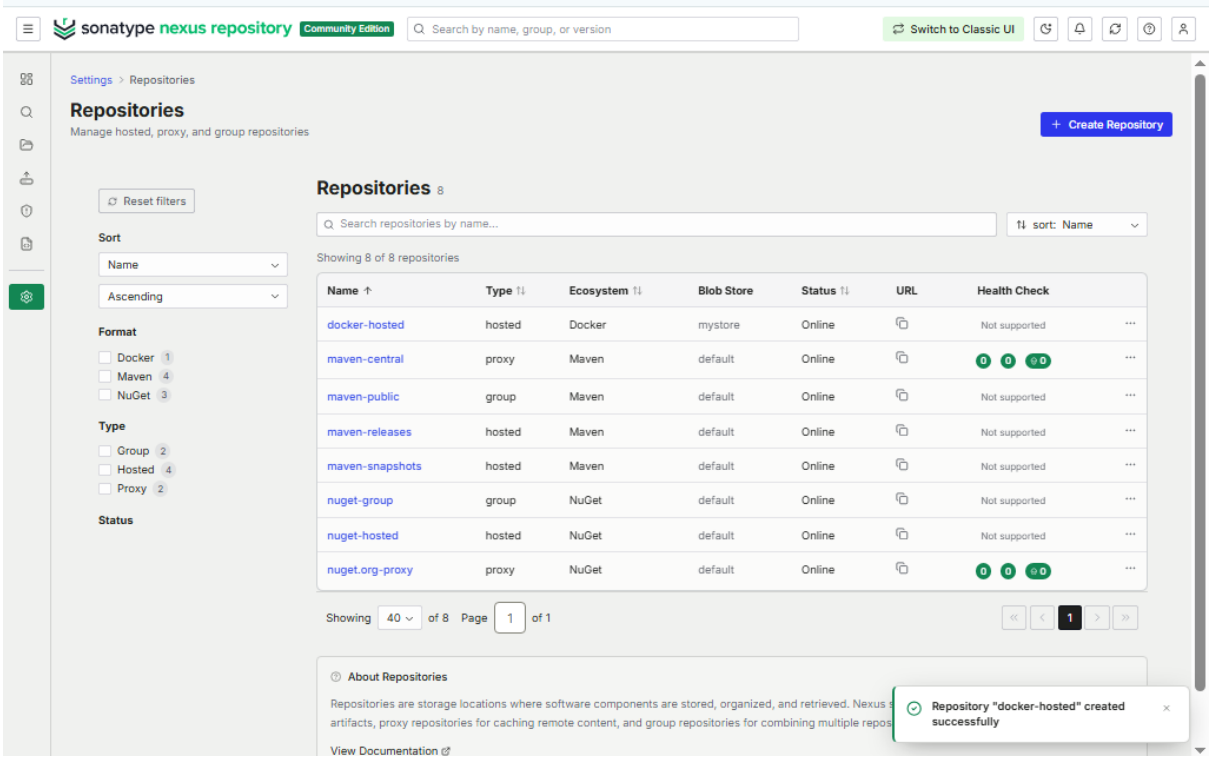
nexus_blob_directories.png



19. Task 6: docker-hosted repository

docker-hosted (hosted, Docker, Online) using the mystore blob store.

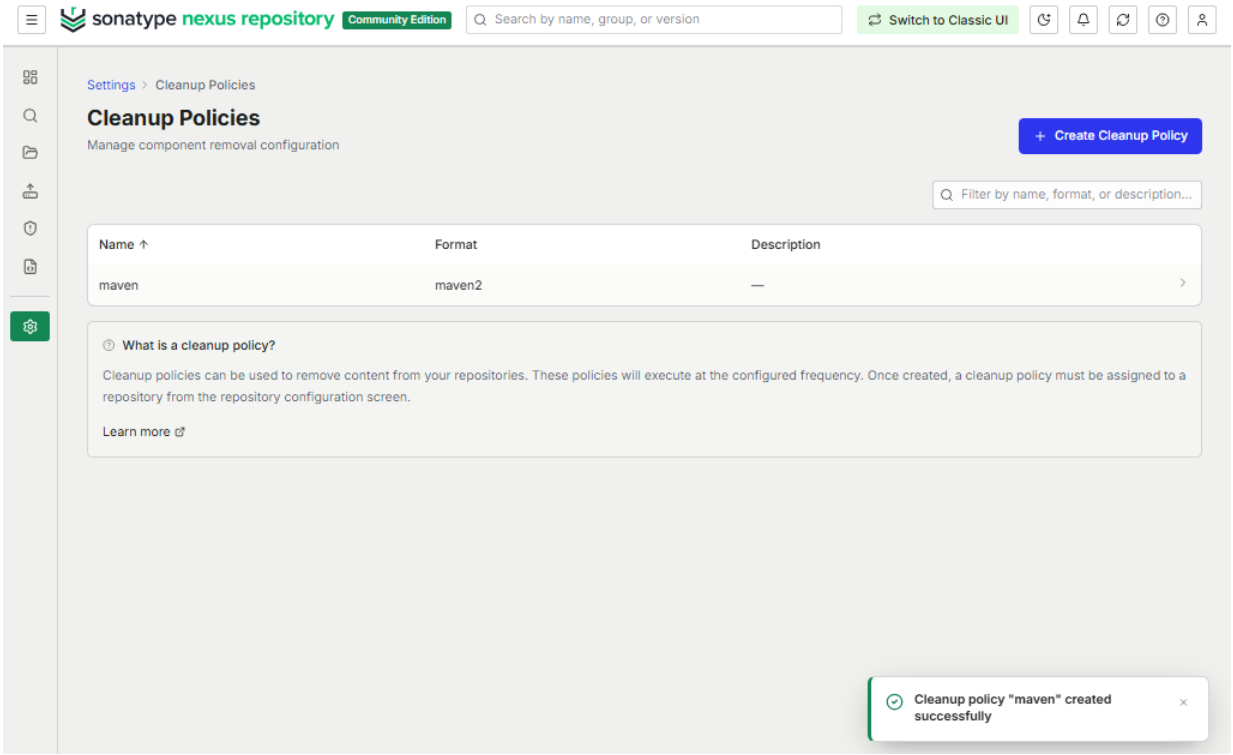
nexus_docker_hosted_repository.png



20. Task 7: cleanup policy 'maven' created

The maven cleanup policy (maven2) is created.

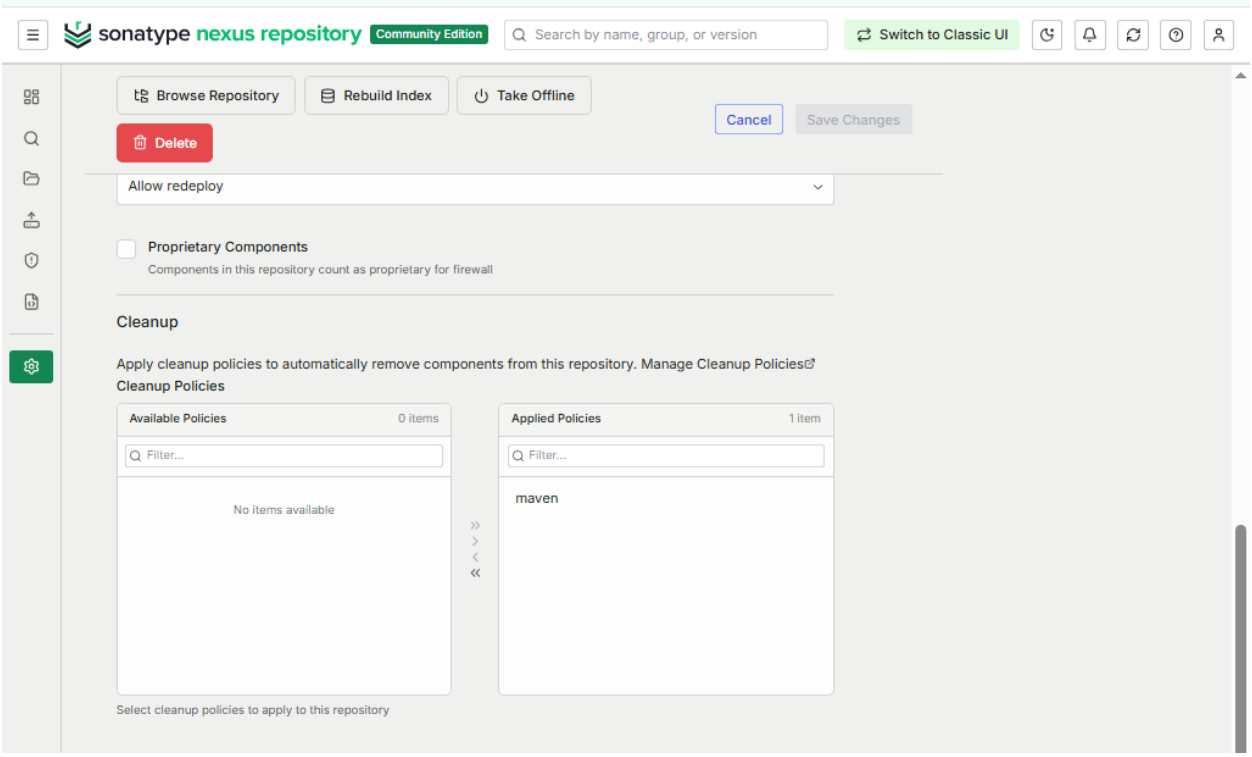
nexus_cleanup_policy_created.png



21. Task 7: maven policy applied to maven-snapshots

maven appears in the Applied Policies list of maven-snapshots.

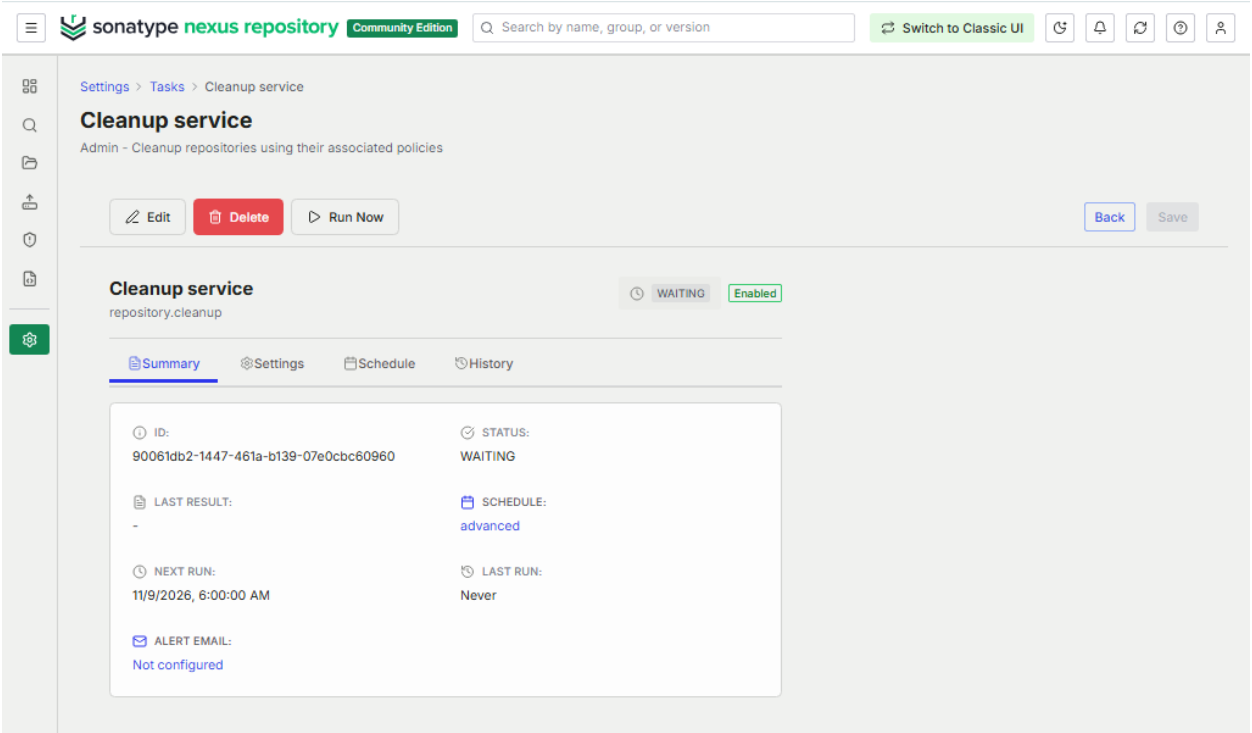
nexus_cleanup_policy_attached.png



22. Task 7: Cleanup service waiting

Cleanup service: status WAITING, next run 6:00 AM.

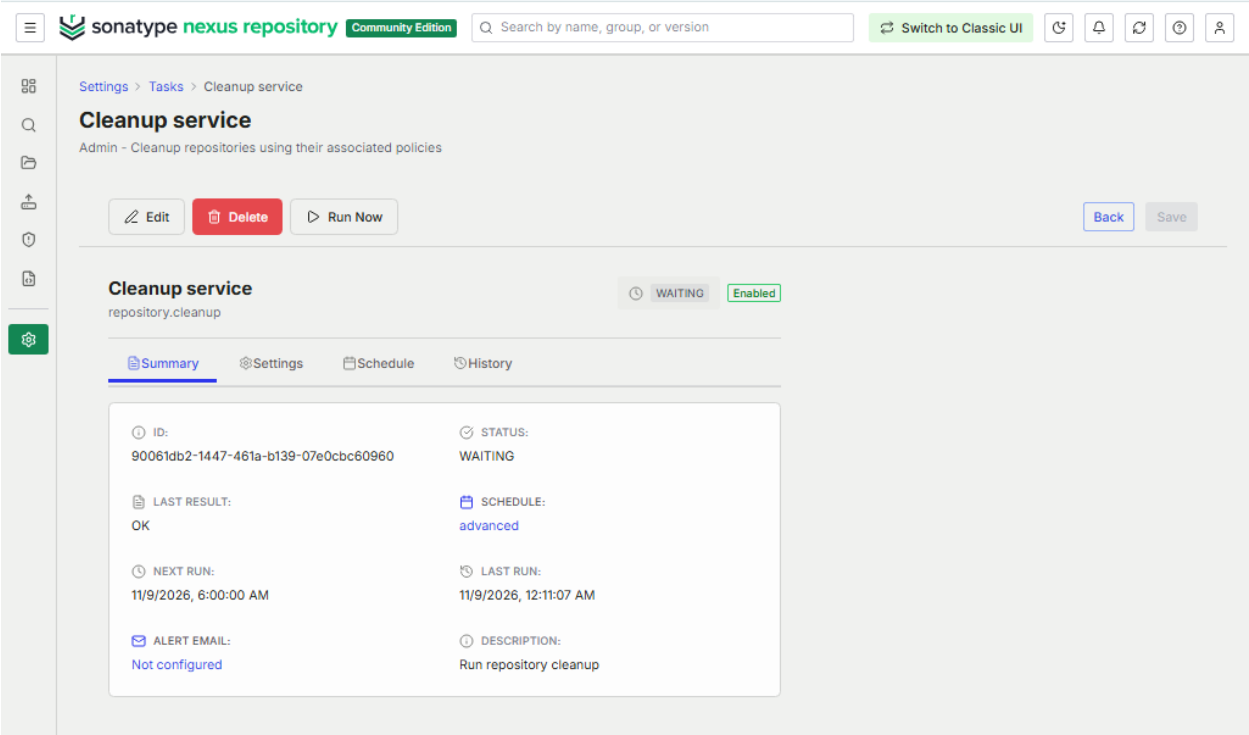
nexus_cleanup_task_waiting.png



23. Task 7: Cleanup service completed

After Run Now: Last result OK with an updated Last run time.

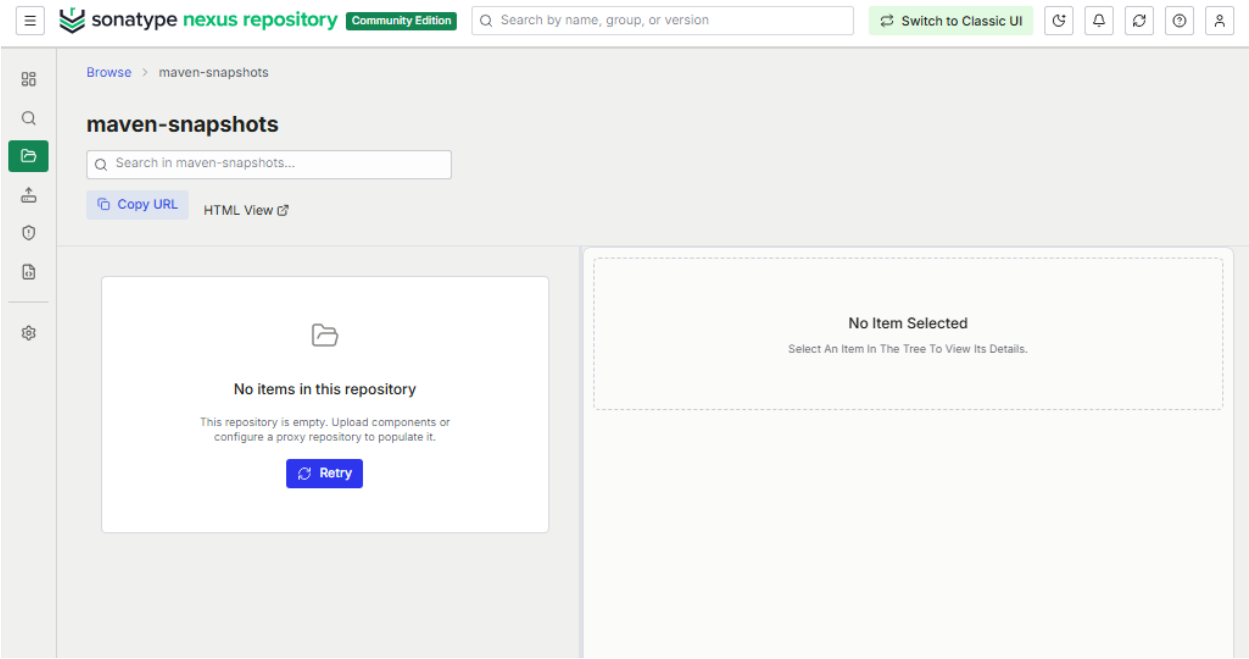
nexus_cleanup_task_completed.png



24. Task 7: maven-snapshots empty after cleanup

maven-snapshots shows no items after the cleanup service ran.

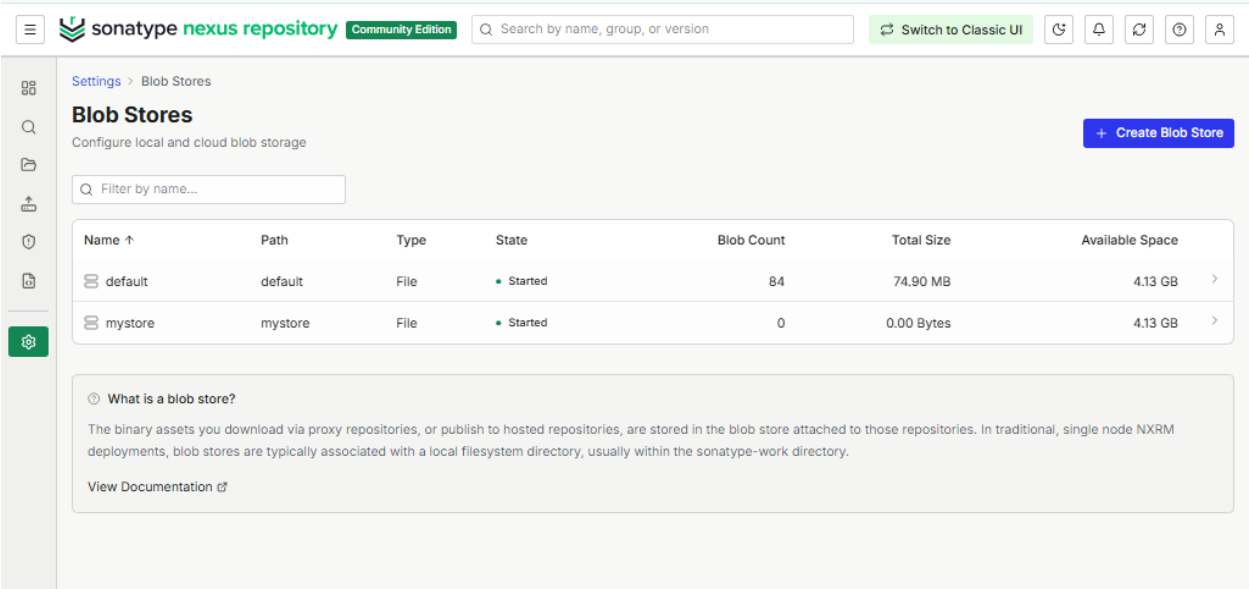
nexus_maven_snapshots_empty.png



25. Task 7: default blob store before compaction

default blob store: blob count, total size and available space before compaction.

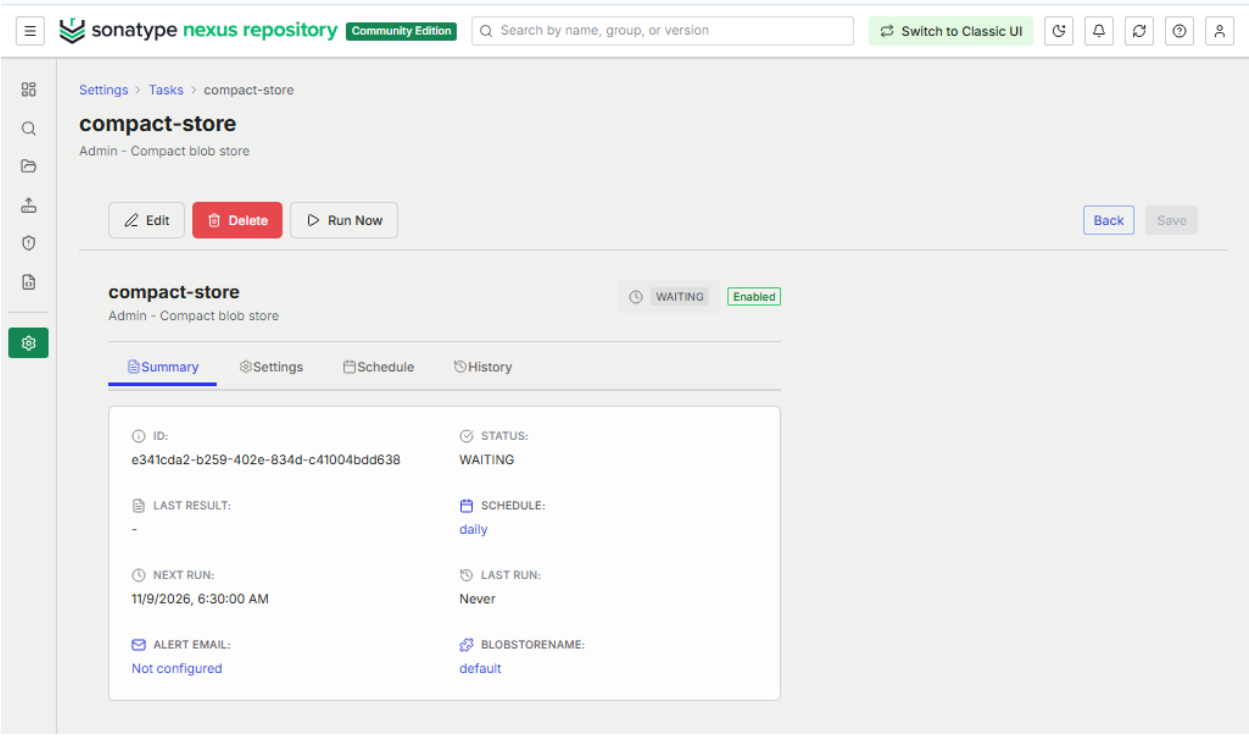
nexus_blob_store_before_compaction.png



26. Task 7: compact-store task configured

compact-store (Admin - Compact blob store): blob store default, daily schedule, next run 6:30 AM.

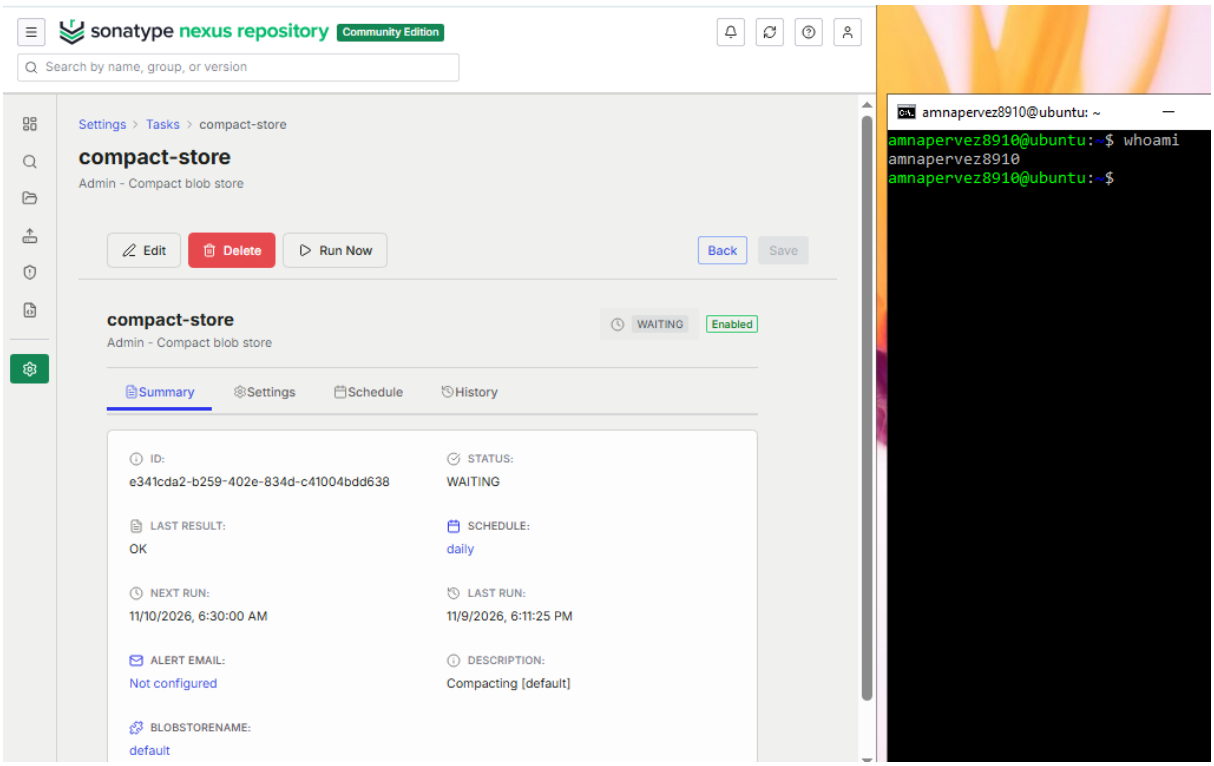
nexus_compact_task_configured.png



27. Task 7: compact-store completed

After Run Now: Last result OK with an updated Last run time; terminal identity visible.

nexus_compact_task_completed.png



28. Task 7: default blob store after compaction

default blob store after compaction with updated blob count, size and available space.

nexus_blob_store_after_compaction.png

